



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

DANG DUC LOC
May 5, 2026



Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary: Falcon 9 First-Stage Landing Prediction

This project provides a comprehensive end-to-end data science analysis of SpaceX's Falcon 9 launch history, fully documented in the [Falcon-9-first-stage-landing-prediction GitHub repository](#). The objective was to uncover the factors driving booster reusability through a structured workflow encompassing data collection via API and web scraping, rigorous data wrangling, and multi-faceted analytical modeling.

Methodology

The project followed an integrated data science pipeline:

- **Data Foundation:** Raw data was cleaned and structured to facilitate feature engineering.
- **Exploratory Data Analysis (EDA):** Insights were extracted using SQL-based querying and visual exploratory techniques, including scatter, bar, and line charts.
- **Interactive Analytics:** Geographical insights were mapped via Folium, while a Plotly Dash application was developed to provide users with a dynamic interface for filtering launch success by payload and site.
- **Predictive Modeling:** A suite of classification algorithms—including Logistic Regression, SVM, Decision Trees, and KNN—was built and optimized via GridSearchCV to forecast landing outcomes.

Key Findings

The analysis confirms a clear trajectory of technological maturity in SpaceX's launch program. EDA revealed a strong positive correlation between time and success rates, alongside critical dependencies between target orbits, booster iterations, and payload capacity. SQL and interactive visualizations demonstrated that while launch sites are strategically located for safety and logistics, the "Block 5" era boosters have successfully mitigated the risks historically associated with high-mass payloads.

Finally, the predictive analysis identified Logistic Regression and SVM as the most robust models, both achieving an 83.3% accuracy rate. These results validate that landing outcomes are highly predictable based on flight parameters, providing a data-driven basis for understanding the economic advantages of SpaceX's reusable flight architecture.

Introduction

Project Background and Context

SpaceX has revolutionized the space industry with its reusable rocket technology, significantly reducing the cost of launching payloads into orbit. Analyzing the vast amount of data generated from its numerous launches provides a valuable opportunity to understand the critical factors that contribute to a successful mission. This project leverages data analytics techniques to delve into SpaceX's historical launch data, aiming to extract actionable insights that can inform future operations and decision-making.

Problems to Investigate

The primary goal of this analysis is to answer several key questions related to SpaceX launch outcomes:

- What is the relationship between launch site and mission success?
- How does payload mass influence the outcome of a launch?
- Are there specific orbital destinations that have higher success rates?
- Is there a discernible trend in launch success over the years?
- Can we build a reliable classification model to predict whether a future launch will be successful based on available pre-launch data?

Section 1

Methodology

Methodology

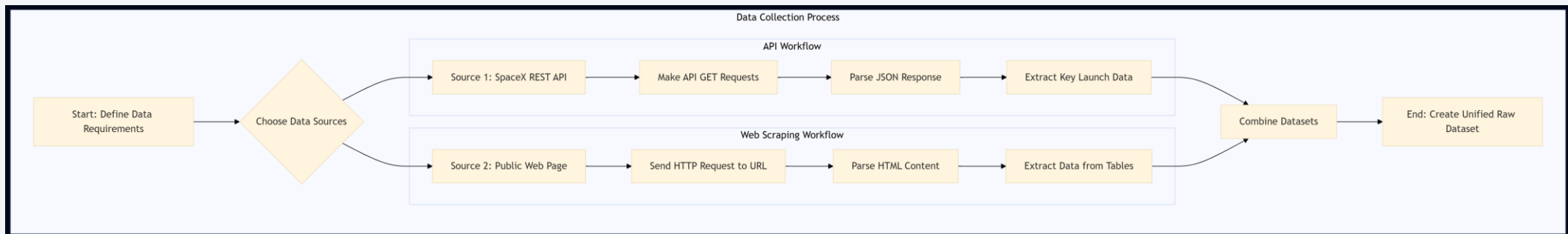
Executive Summary

- Data collection methodology:
 - Gathering data from the SpaceX API and through web scraping
- Perform data wrangling
 - Cleaning, structuring, and preparing the collected raw data for analysis
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
 - Building, tuning, and evaluating various classification models to predict launch outcomes and identifying the best-performing model.

Data Collection

The key phrases that describe the data collection process are:

- **Primary Data Source:** The core technical and mission-specific data was gathered by making programmatic API calls to the official SpaceX REST API. This provided structured JSON data containing detailed information on launch missions, rocket configurations, payload specifics, and flight outcomes.
- **Secondary Data Source:** To supplement the API data and enrich the dataset, web scraping techniques were employed. This involved extracting tabular data from publicly available web pages (such as Wikipedia) to gather additional historical context and information not available through the API.
- **Unified Raw Dataset:** The data streams from both the API and web scraping were then integrated to form a single, unified raw dataset. This combined dataset served as the foundational input for the subsequent Data Wrangling stage. You need to present your data collection process use key phrases and flowcharts



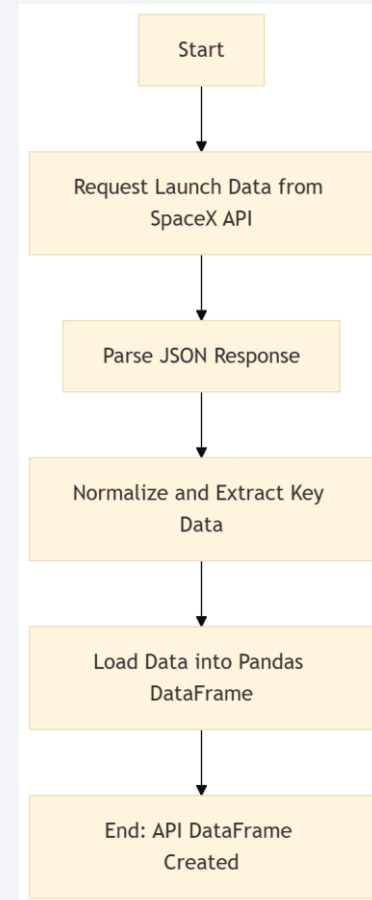
Data Collection – SpaceX API

The process for collecting data from the SpaceX API can be described using the following key phrases:

- **API Endpoint Identification:** The process began by identifying the specific API endpoint (/launches) that provides detailed information for all Falcon 9 missions.
- **HTTP GET Request:** Using Python's requests library, an HTTP GET request was sent to the identified endpoint to retrieve the launch data.
- **JSON Response Parsing:** The API responded with a list of launches formatted as a JSON object. This nested JSON structure was programmatically parsed to access the required data fields for each mission.
- **Data Normalization:** Since the raw JSON data was nested (e.g., rocket and payload information were objects within the main launch object), it was normalized or "flattened." This converted the complex structure into a flat, tabular format suitable for analysis.
- **Pandas DataFrame Creation:** The normalized data was then loaded into a Pandas DataFrame, which serves as the primary data structure for the subsequent data wrangling and analysis stages.

GitHub URL of the completed SpaceX API calls notebook:

<https://github.com/aanpanman/Falcon-9-first-stage-landing-prediction/blob/main/1.%20jupyter-labs-spacex-data-collection-api-v2.ipynb>



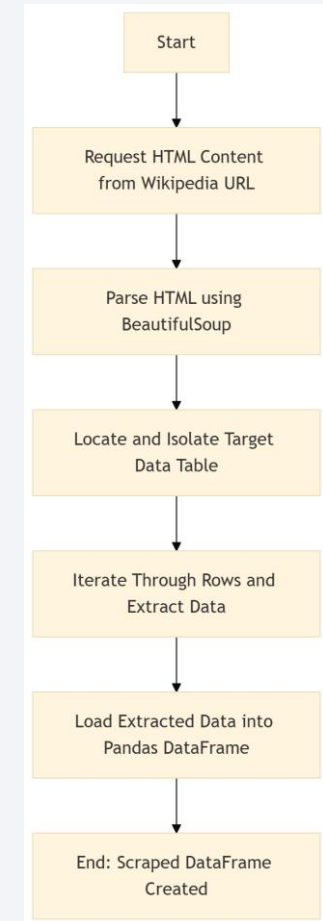
Data Collection - Scraping

The process was executed using the following key steps:

- Target URL Request: An HTTP GET request was sent to the target URL using a Python library to fetch the raw HTML content of the page.
- HTML Parsing: The retrieved HTML content was parsed using the BeautifulSoup library, transforming it into a structured object that could be easily navigated and queried.
- Data Extraction: The parsed HTML object was searched to locate the specific HTML table containing the desired launch information. The data from the table rows and cells was then systematically extracted.
- DataFrame Creation: The extracted data was organized and loaded into a Pandas DataFrame. This ensured the scraped data was in the same tabular format as the API data, allowing for a seamless merge in the data wrangling stage.

GitHub URL of the completed web scraping notebook:

<https://github.com/aanpanman/Falcon-9-first-stage-landing-prediction/blob/main/2.%20jupyter-labs-webscraping.ipynb>



Data Wrangling

Here is the step-by-step description of how the data were processed, highlighting the **key phrases** of the wrangling process:

- Data Ingestion & Profiling: Loaded the raw dataset and assessed missing values and data types to understand feature structures.
- Exploratory Data Analysis (EDA): Calculated occurrences and traffic distributions across Launch Sites and targeted Orbits.
- Outcome Evaluation: Reviewed string-based mission outcomes (e.g., True Ocean, False RTLS, None ASDS).
- Feature Engineering (Target Label): Created a binary Class label by mapping successful landings to 1 and unsuccessful/no-attempt landings to 0.
- Baseline Metric Calculation: Computed the overall Falcon 9 first-stage landing success rate (~66.6%).
- Export: Saved the processed dataset with the new target variable for the SQL analysis and Machine Learning phases.

GitHub URL of the completed data wrangling related notebooks:

<https://github.com/aanpanman/Falcon-9-first-stage-landing-prediction/blob/main/3.%20labs-jupyter-spacex-Data%20wrangling-v2.ipynb>



EDA with Data Visualization

Flight History Trends (Scatter Plots):

- Plotted: Flight Number vs. Payload Mass, Launch Site, and Orbit.
- Why: To track the evolution of SpaceX operations over time and identify how growing experience impacted successful landing clusters.

Payload Constraints (Scatter Plots):

- Plotted: Payload Mass vs. Launch Site and Orbit.
- Why: To evaluate weight limits across different facilities and destinations and determine if heavier payloads face higher failure rates.

Orbit Performance (Bar Chart):

- Plotted: Average Success Rate by Orbit Type.
- Why: To directly compare landing reliability across mission profiles, instantly highlighting the safest vs. most difficult orbits.

Historical Progress (Line Chart):

- Plotted: Landing Success Rate vs. Year.
- Why: To clearly illustrate the steady, year-over-year improvement in SpaceX's booster recovery technology.

GitHub URL of the completed EDA with data visualization notebook:

<https://github.com/aanpanman/Falcon-9-first-stage-landing-prediction/blob/main/5.%20jupyter-labs-eda-dataviz-v2.ipynb>

EDA with SQL

Here is a summary of the SQL queries performed during the Exploratory Data Analysis

- Database Setup & Cleaning: Dropped any existing SPACEXTABLE and created a new one by copying data from the loaded pandas dataframe (SPACEXTBL), specifically filtering out rows where the Date was null.
- Unique Launch Sites (Task 1): Used SELECT DISTINCT to retrieve all the unique launch facility names from the dataset.
- Launch Site Filtering (Task 2): Queried the first 5 records where the launch site name begins with the string 'CCA' using the LIKE 'CCA%' operator combined with LIMIT 5.
- Total Payload Mass by Customer (Task 3): Calculated the total overall payload mass carried for the customer 'NASA (CRS)' using the SUM() aggregate function on the payload mass column.
- Average Payload Mass by Booster (Task 4): Computed the average payload mass carried specifically by the 'F9 v1.1' booster version using the AVG() function.
- First Successful Ground Pad Landing (Task 5): Found the date of the very first successful ground pad landing by using the MIN() function on the Date column where the landing outcome was 'Success (ground pad)'.
- Booster Search by Payload and Outcome (Task 6): Listed distinct booster versions that had a successful drone ship landing and carried a payload mass between 4,000 and 6,000 kg using the BETWEEN operator.
- Mission Outcomes Count (Task 7): Counted the total number of missions for each specific mission outcome (e.g., Success, Failure) using COUNT(*) and the GROUP BY clause.
- Maximum Payload Mass Boosters (Task 8): Used a subquery (SELECT MAX(PAYLOAD_MASS_KG_)) to identify and list the specific booster versions that carried the heaviest payload recorded in the entire dataset.
- 2015 Drone Ship Failures (Task 9): Filtered for 'Failure (drone ship)' outcomes specifically in the year 2015. Because SQLite doesn't natively support month names, the query used the substr() function to extract the year and month from the Date string and a CASE statement to convert the month numbers into month names.
- Landing Outcome Ranking (Task 10): Counted the occurrences of each landing outcome within a specific date range ('2010-06-04' to '2017-03-20'), grouped them by outcome, and ordered the results in descending frequency using ORDER BY ... DESC.

GitHub URL of the completed EDA with SQL notebook: https://github.com/aanpanman/Falcon-9-first-stage-landing-prediction/blob/main/4.%20jupyter-labs-eda-sql-coursera_sqlite.ipynb

Build an Interactive Map with Folium

Here is a summary of the map objects created and added to a folium map:

- Launch Site Circles & Text Markers:
 - What: folium.Circle objects and custom text labels (DivIcon) placed at the exact coordinates of each facility (e.g., KSC LC-39A, VAFB SLC-4E).
 - Why: To establish the baseline geographic layout and pinpoint where SpaceX operates its primary launch sites.
- Mission Outcome Marker Clusters:
 - What: A MarkerCluster grouping individual flight markers, color-coded by landing outcome (Green = Success, Red = Failure).
 - Why: To visually assess the volume of traffic and the historical success/failure ratio at each site simultaneously without cluttering the map.
- Proximity Lines (PolyLine) & Distance Labels:
 - What: Lines drawn connecting launch sites to specific Points of Interest (coastlines, highways, railways, and cities), alongside markers displaying the calculated distance (Haversine formula).
 - Why: To analyze the strategic placement of launch sites regarding logistics (transporting heavy rockets via rail/highway) and safety (proximity to oceans for safe staging/aborts and distance from dense populations).

GitHub URL of the completed interactive map with Folium map:

<https://github.com/aanpanman/Falcon-9-first-stage-landing-prediction/blob/main/6.%20lab-jupyter-launch-site-location-v2.ipynb>

Build a Dashboard with Plotly Dash

Here is a summary of the dashboard:

User Interactions (Inputs)

- Launch Site Dropdown: Allows users to select 'All Sites' or an individual launch facility.
 - Why: To dynamically filter the dashboard data, enabling both high-level global analysis and granular, site-specific evaluations.
- Payload Mass Range Slider: Lets users select a specific payload weight range (0 to 10,000 kg).
 - Why: To isolate specific mass categories and test the hypothesis that heavier payloads might impact the likelihood of a successful landing.

Visualizations (Outputs)

- Interactive Pie Chart: Displays the proportion of total successful launches per site (Global view) or the Success vs. Failure ratio for a single facility (Site view).
 - Why: To instantly visualize the historical reliability of individual launch pads and see which sites contribute most to overall success.
- Color-Coded Scatter Plot: Plots Payload Mass vs. Landing Outcome (0 or 1), with points categorized by Booster Version.
 - Why: To analyze complex correlations between payload weight constraints, booster rocket types, and landing success within user-defined parameters.

GitHub URL of the completed Plotly Dash lab:

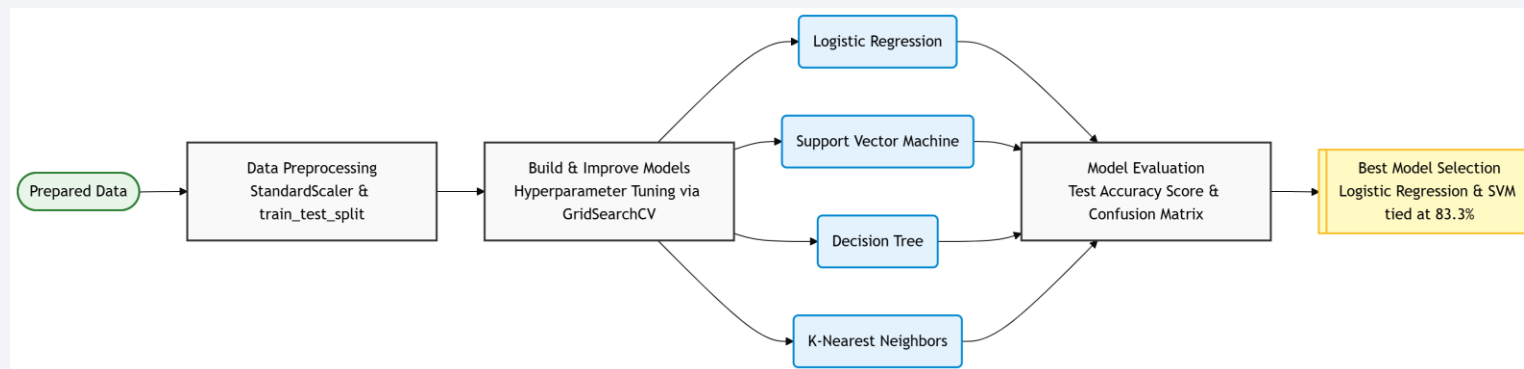
https://github.com/aanpanman/Falcon-9-first-stage-landing-prediction/blob/main/spacex_dash_app.py

Predictive Analysis (Classification)

Machine Learning Prediction Pipeline:

- Build (Data Preparation): Standardized the feature dataset using StandardScaler and partitioned it into 80/20 training and testing sets using train_test_split.
- Improve (Hyperparameter Tuning): Built four distinct classification algorithms (Logistic Regression, SVM, Decision Tree, K-Nearest Neighbors) and improved them by finding optimal parameters using GridSearchCV (with 10-fold cross-validation).
- Evaluate (Performance Metrics): Assessed each model's predictive capability by calculating the test set Accuracy Score (.score()) and plotting Confusion Matrices to visualize True vs. False Positives/Negatives.
- Best Performing Model: Compared the final test accuracies to identify the strongest predictor. Logistic Regression and Support Vector Machine (SVM) tied as the best performing models, both achieving an accuracy of 83.3% on the test data.

GitHub URL of the completed predictive analysis lab: <https://github.com/aanpanman/Falcon-9-first-stage-landing-prediction/blob/main/7.%20SpaceX-Machine-Learning-Prediction-Part-5-v1.ipynb>



Key Findings & Prediction Results

Exploratory Data Analysis (EDA) Results:

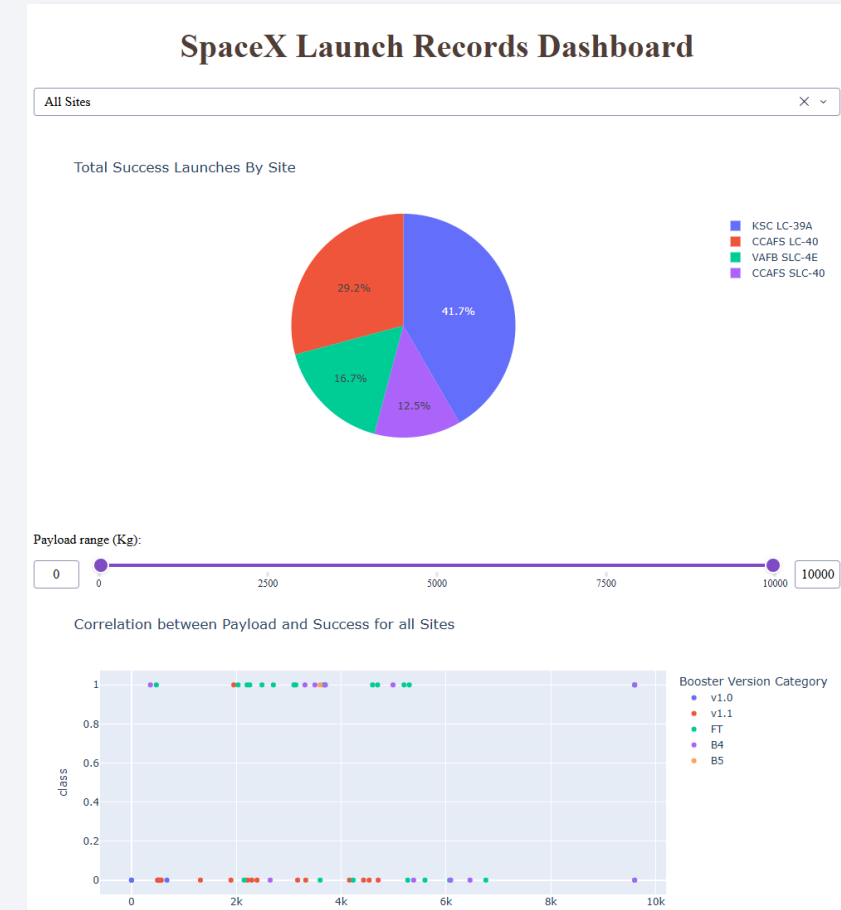
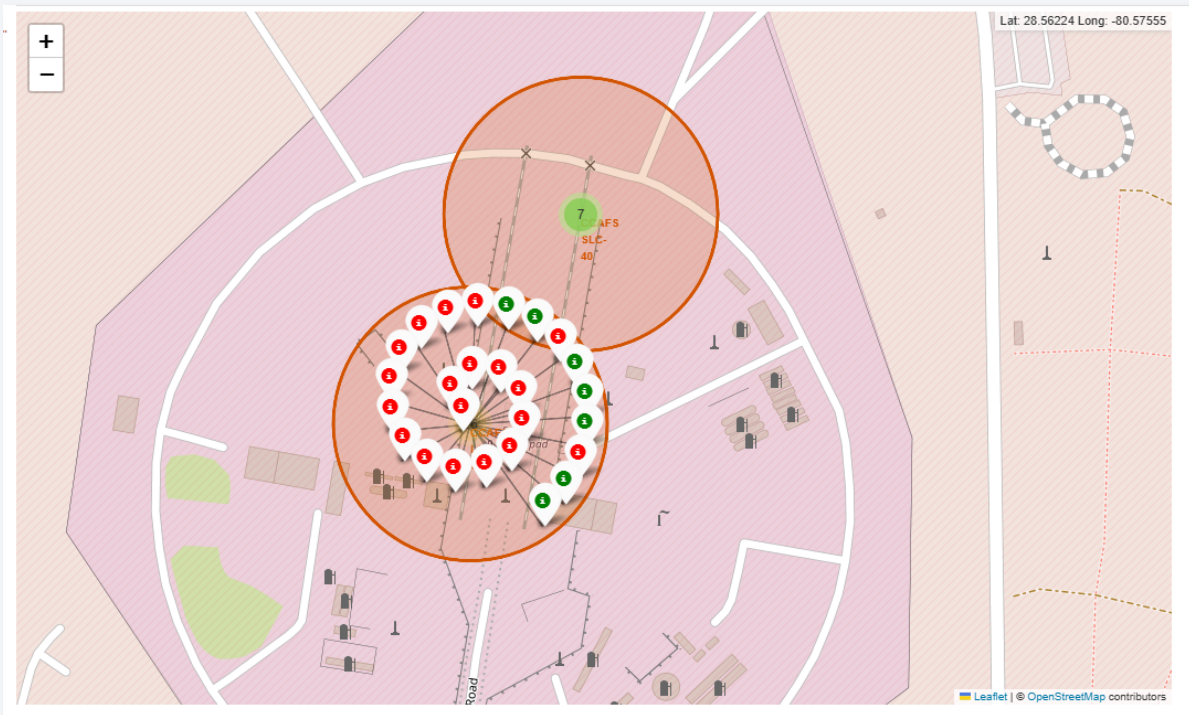
- **Historical Progress:** Landing success rates have shown a steady, year-over-year improvement as SpaceX refined its technology.
- **Mission Profile Impact:** Lighter payload masses and specific target orbits (e.g., SSO, HEO, GEO, ES-L1) heavily correlate with higher landing success.
- **Strategic Logistics:** Launch facilities are strategically positioned near coastlines and transport networks (rail/highway) for safety and heavy-payload logistics.
- **Site Performance:** Launch success ratios vary by facility, with heavily utilized sites like CCAFS SLC-40 having the most attempts, while KSC LC-39A shows higher overall reliability.

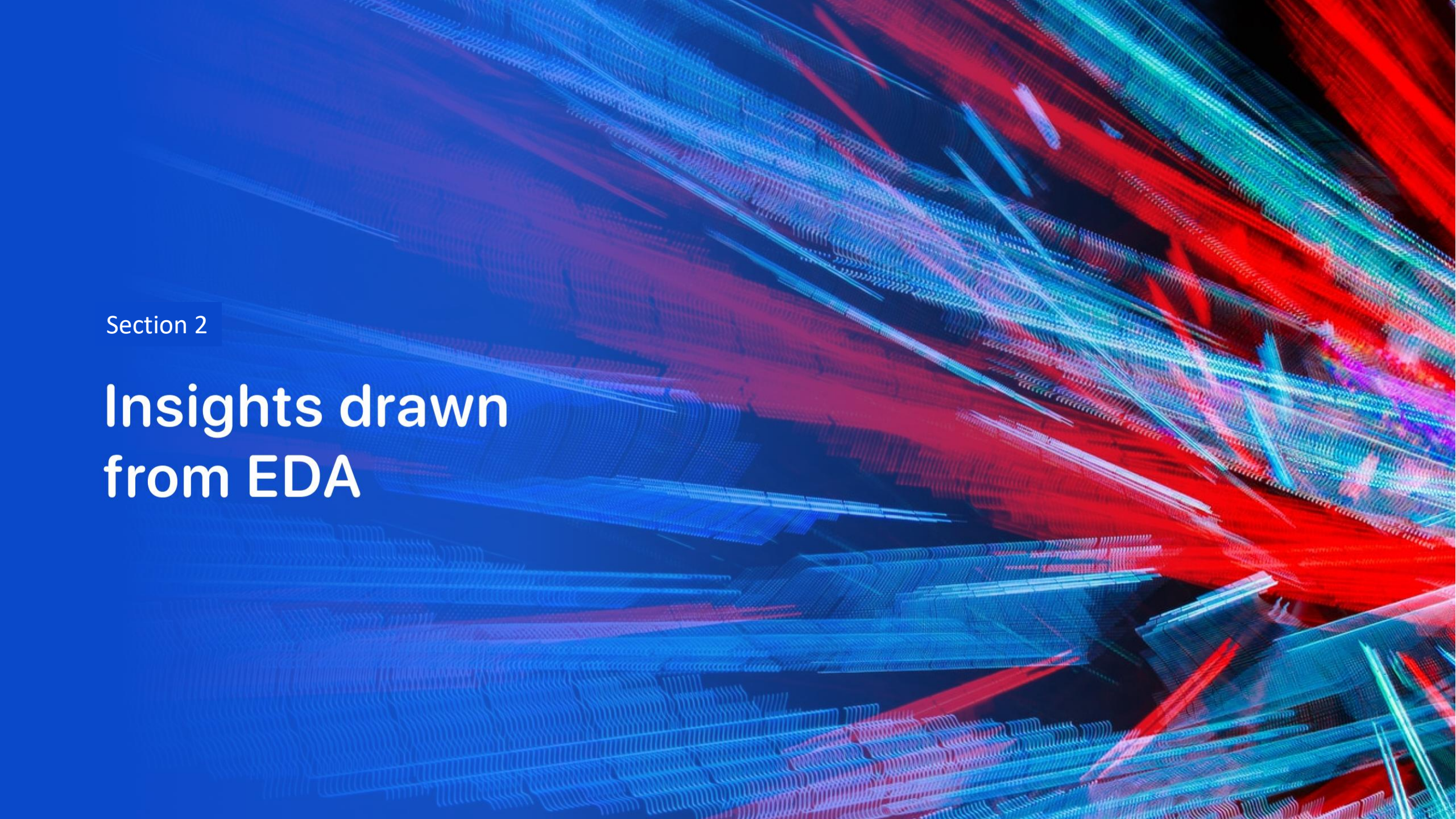
Predictive Analysis (Machine Learning) Results:

- **Overall Performance:** Flight parameters are strong indicators of landing outcomes. After standardizing the data, all tested models performed exceptionally well.
- **Hyperparameter Tuning:** Using GridSearchCV optimized the models, effectively minimizing false positives and false negatives (verified via Confusion Matrices).
- **Best Performing Models:** Logistic Regression and Support Vector Machine (SVM) were identified as the top models, both achieving an impressive 83.3% accuracy on the test dataset.
- **Conclusion:** Falcon 9 first-stage landing outcomes can be reliably predicted, allowing for highly accurate launch cost estimations

Deliverables

Interactive analytics demo in screenshots

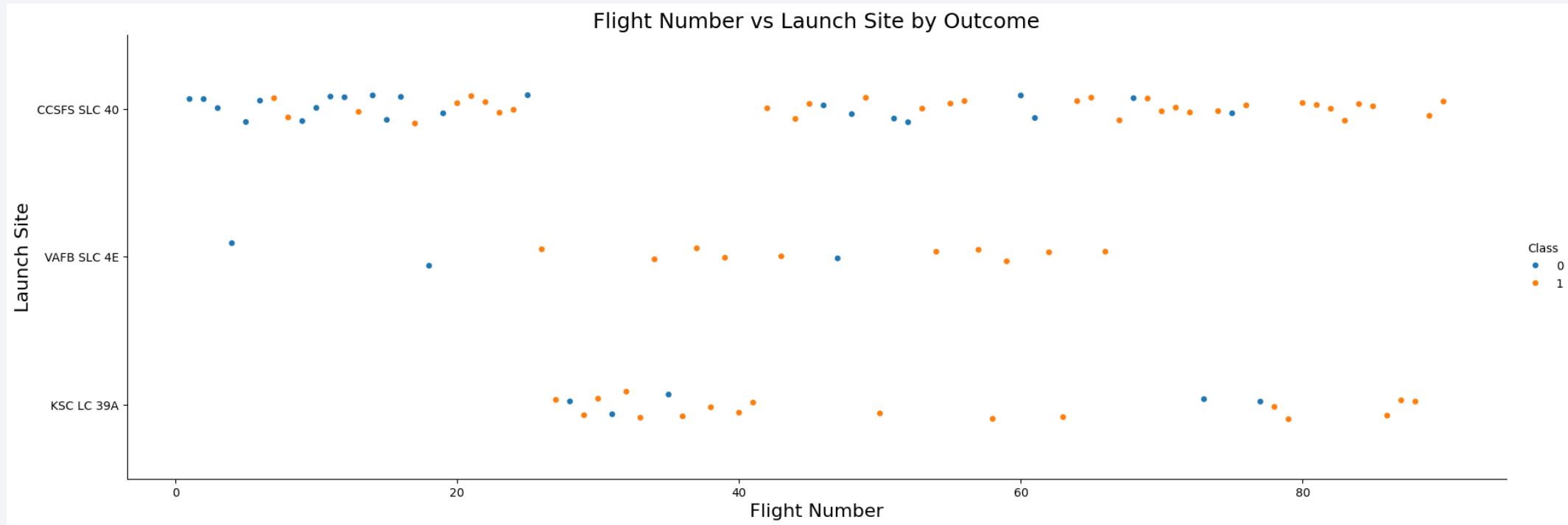


The background of the slide is an abstract composition. It features a dark blue gradient on the left side, which transitions into a complex pattern of diagonal streaks and lines in shades of blue, red, and teal on the right. These streaks have a textured, almost woven appearance, suggesting a digital or data-driven theme. The overall effect is dynamic and modern.

Section 2

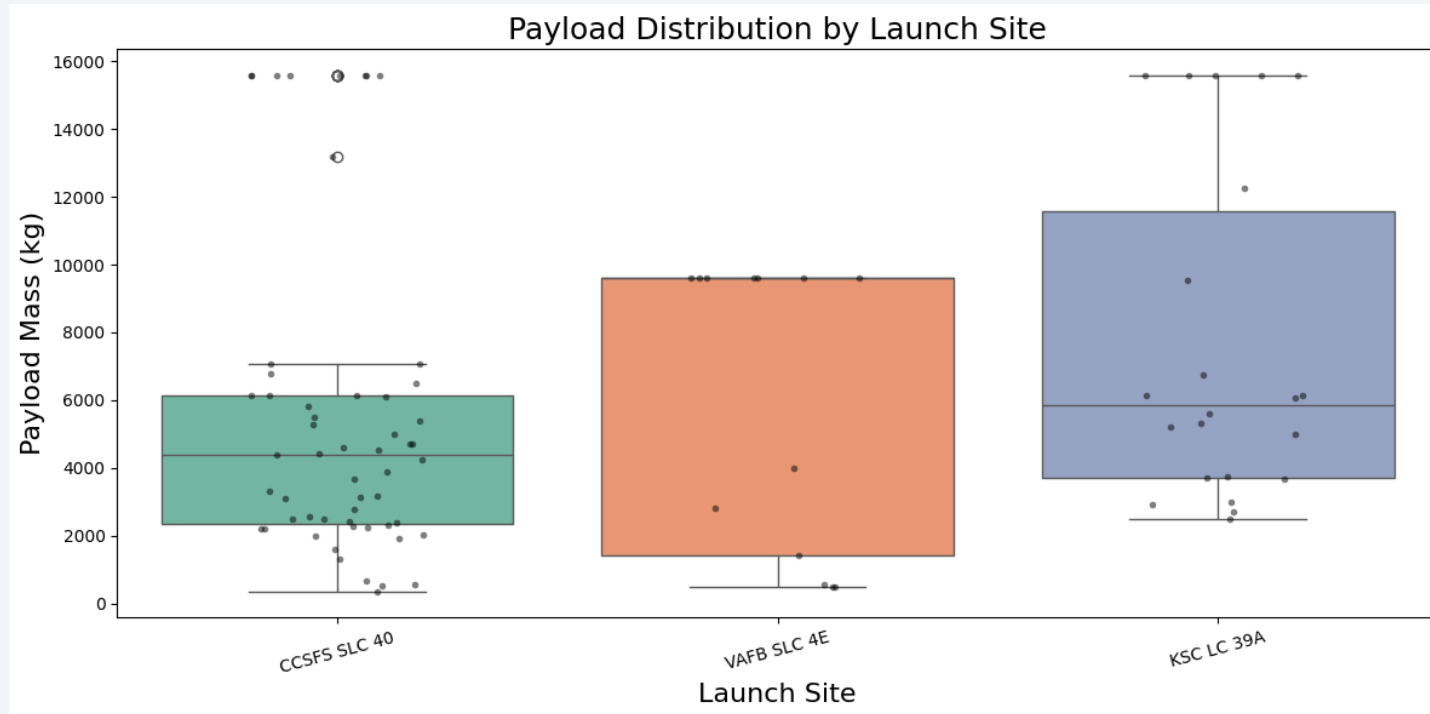
Insights drawn from EDA

Flight Number vs. Launch Site



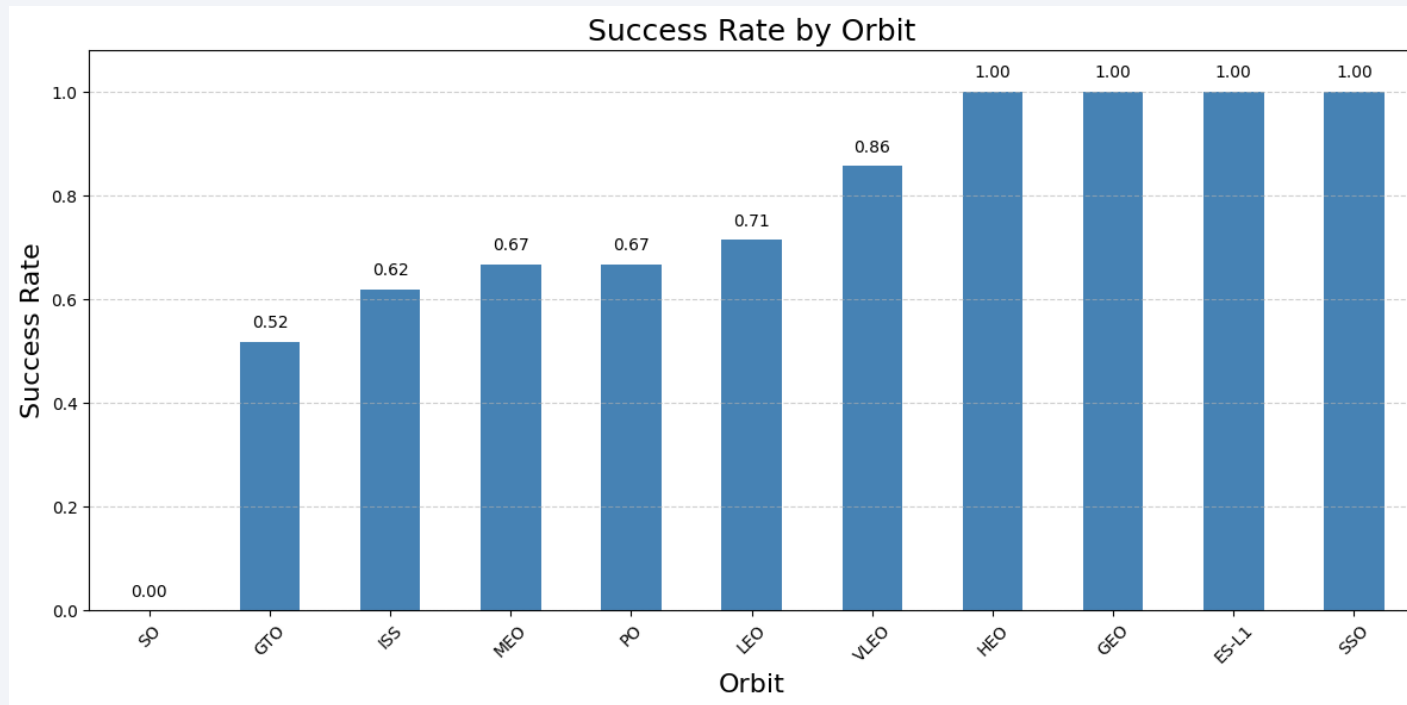
- Relation: As flight numbers increase (representing the passage of time and gained experience), there is a clear shift in site usage and landing success.
- Observation: CCAFS SLC 40 was heavily used for early test flights and lower flight numbers, which had more mixed landing outcomes. As flight numbers increased, SpaceX activated KSC LC 39A and VAFB SLC 4E. Most importantly, higher flight numbers show a much higher density of successful landings (Class 1) across all launch sites, indicating overall technological improvement regardless of the facility used.

Payload vs. Launch Site



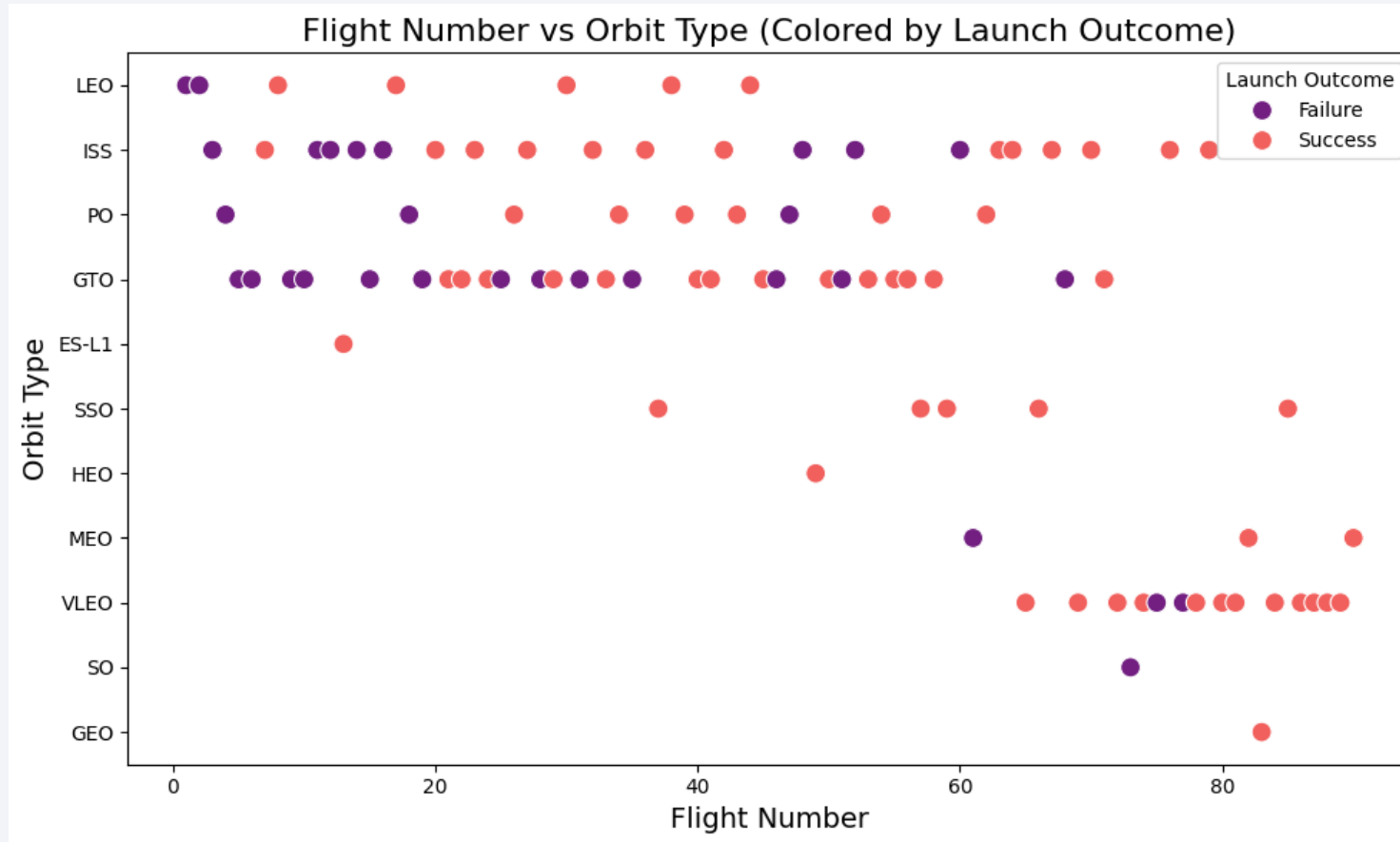
- Relation: Different launch sites are utilized for different payload mass capacities.
- Observation: VAFB SLC 4E is predominantly used for lighter payloads (mostly under 10,000 kg). In contrast, CCAFS SLC 40 and KSC LC 39A are the workhorse sites, handling a much broader range of payload masses, including the heaviest payloads (up to ~15,600 kg). While heavier payloads might seem harder to land, KSC and CCAFS still maintain high success rates even with maximum payload masses.

Success Rate vs. Orbit Type



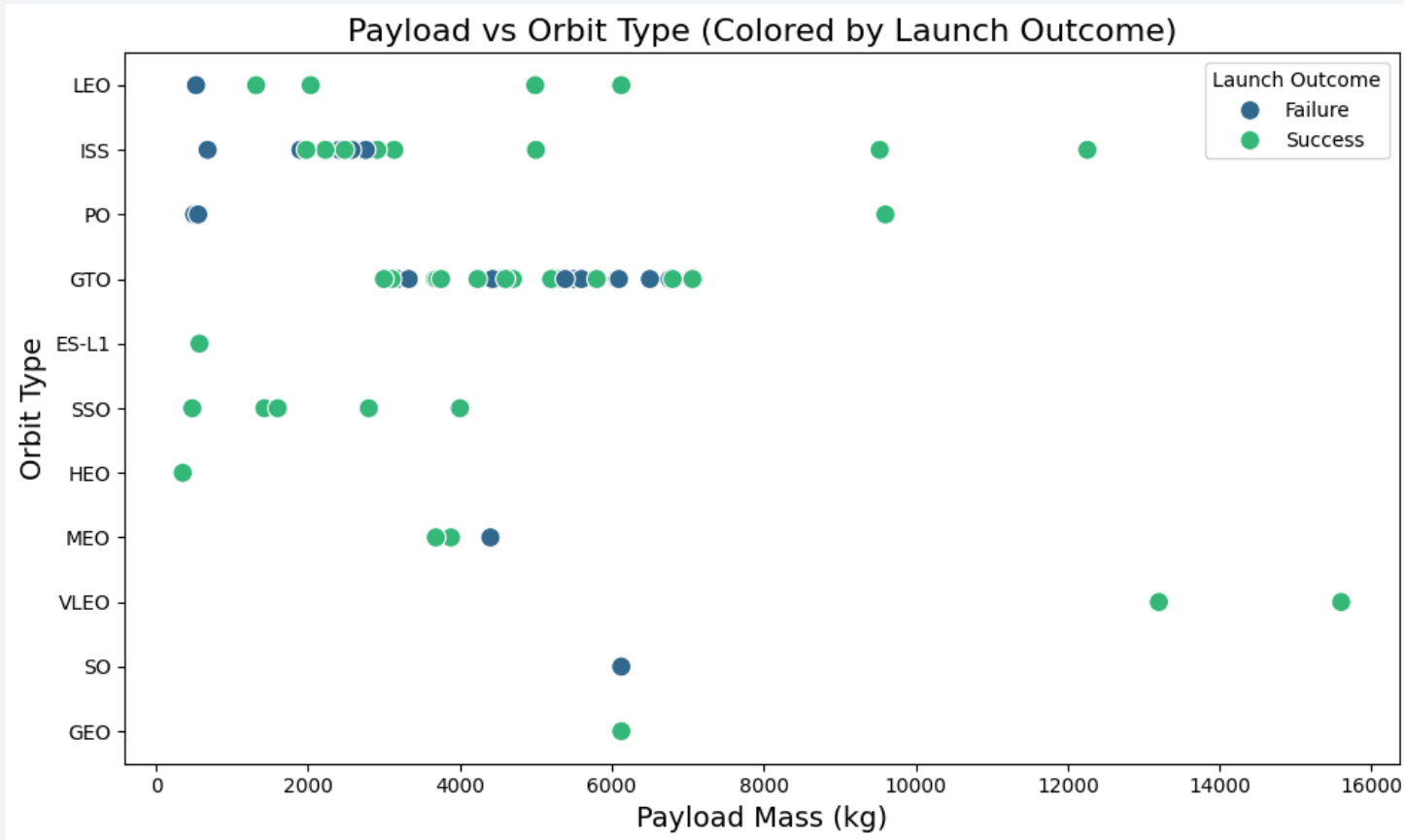
- Relation: The target orbit has a strong correlation with the likelihood of a successful booster landing.
- Observation: Certain orbits are highly reliable for booster recovery. Orbits like ES-L1, GEO, HEO, and SSO show a 100% success rate in this dataset. High-traffic orbits like ISS and VLEO also show very strong success rates. Conversely, orbits like SO (Sun-synchronous Orbit) show a 0% success rate, and GTO (Geostationary Transfer Orbit) shows a relatively lower success rate (~50%), as sending rockets deeper into space requires more fuel, leaving less fuel for a controlled descent.

Flight Number vs. Orbit Type



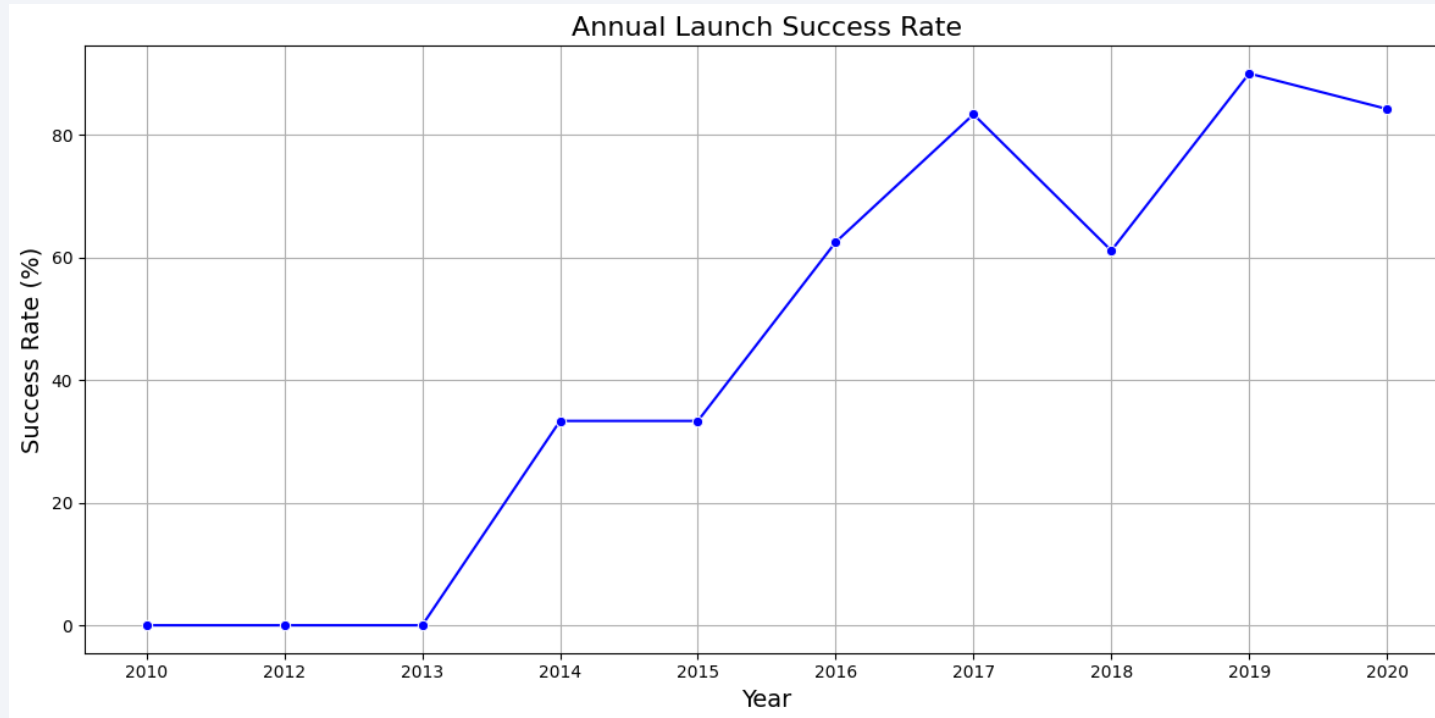
- Relation: SpaceX's mission profiles and targeted destinations have evolved alongside their flight experience.
- Observation: In earlier flights (lower flight numbers), SpaceX frequently targeted the ISS, LEO, and GTO. In later flights (higher flight numbers), there is a massive cluster of missions directed to VLEO (Very Low Earth Orbit). This later cluster corresponds to the deployment of SpaceX's Starlink satellite constellation, and nearly all of these recent VLEO missions resulted in successful booster landings.

Payload vs. Orbit Type



- Relation: Payload mass is heavily dictated by the destination orbit.
- Observation: Heavy payloads (around 15,000 kg) are almost exclusively sent to VLEO (Starlink missions) and ISS. GTO and PO (Polar Orbit) typically handle mid-range payload weights. Orbits like SSO, HEO, and ES-L1 are generally associated with much lighter payloads. The visualizations show that despite the massive weight of VLEO payloads, the landing success rate remains extremely high.

Launch Success Yearly Trend



- Relation: Time (Years) is directly and positively correlated with the average landing success rate.
- Observation: The line chart shows a dramatic technological progression. From 2010 to 2013, the success rate was 0% (as SpaceX was still testing or not attempting landings). Starting in 2014, there is a sharp and steady upward trend. By 2019 and 2020, the average success rate stabilized between 80% and 90%+. This illustrates SpaceX's successful mastery of first-stage booster recovery over the decade.

All Launch Site Names

Display the names of the unique launch sites in the space mission

In [11]:

```
%sql SELECT DISTINCT Launch_Site FROM SPACEXTABLE
```

```
* sqlite:///my_data1.db
```

Done.

Out[11]:

Launch_Site

CCAFS LC-40

VAFB SLC-4E

KSC LC-39A

CCAFS SLC-40

Launch Site Names Begin with 'CCA'

Display 5 records where launch sites begin with the string 'CCA'

```
%sql SELECT * FROM SPACEXTABLE WHERE Launch_Site LIKE 'CCA%' LIMIT 5
```

* sqlite:///my_data1.db

Done.

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Total Payload Mass

Task 3

Display the total payload mass carried by boosters launched by NASA (CRS)

```
%sql SELECT SUM(PAYLOAD_MASS__KG_) AS TOTAL_PAYLOAD_MASS FROM SPACEXTABLE WHERE Customer = 'NASA (CRS)'
```

```
* sqlite:///my_data1.db
```

Done.

<u>TOTAL_PAYLOAD_MASS</u>

45596

Average Payload Mass by F9 v1.1

Display average payload mass carried by booster version F9 v1.1

```
%sql SELECT AVG(PAYLOAD_MASS__KG_) AS AVERAGE_PAYLOAD_MASS FROM SPACEXTABLE WHERE Booster_Version = 'F9 v1.1'
```

```
* sqlite:///my_data1.db
```

```
Done.
```

```
* sqlite:///my_data1.db
```

```
Done.
```

AVERAGE_PAYLOAD_MASS

2928.4

First Successful Ground Landing Date

```
%sql SELECT MIN(Date) FROM SPACEXTABLE WHERE Landing_Outcome = 'Success (ground pad)'
```

```
* sqlite:///my_data1.db
```

```
Done.
```

```
* sqlite:///my_data1.db
```

```
Done.
```

<u>MIN(Date)</u>

2015-12-22

Successful Drone Ship Landing with Payload between 4000 and 6000

List the names of the boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000

```
%sql SELECT DISTINCT Booster_Version FROM SPACEXTABLE WHERE Landing_Outcome = 'Success (drone ship)' AND PAYLOAD_MASS_KG_ BETWEEN 4000 AND 6000
```

```
* sqlite:///my_data1.db
```

```
Done.
```

```
* sqlite:///my_data1.db
```

```
Done.
```

Booster_Version

F9 FT B1022

F9 FT B1026

F9 FT B1021.2

F9 FT B1031.2

Total Number of Successful and Failure Mission Outcomes

List the total number of successful and failure mission outcomes

```
%sql SELECT Mission_Outcome, COUNT(*) AS NUMBER_OF_MISSIONS FROM SPACEXTABLE GROUP BY Mission_Outcome
```

```
* sqlite:///my_data1.db
```

```
Done.
```

```
* sqlite:///my_data1.db
```

```
Done.
```

Mission_Outcome	NUMBER_OF_MISSIONS
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

Boosters Carried Maximum Payload

List all the booster_versions that have carried the maximum payload mass, using a subquery with a suitable aggregate function.

```
%sql SELECT DISTINCT Booster_Version, PAYLOAD_MASS__KG_ FROM SPACEXTABLE WHERE PAYLOAD_MASS__KG_ = (SELECT MAX(PAYLOAD_MASS__KG_) FROM SPACEXTABLE)
```

```
* sqlite:///my_data1.db
```

Done.

Booster_Version	PAYLOAD_MASS__KG_
F9 B5 B1048.4	15600
F9 B5 B1049.4	15600
F9 B5 B1051.3	15600
F9 B5 B1056.4	15600
F9 B5 B1048.5	15600
F9 B5 B1051.4	15600
F9 B5 B1049.5	15600
F9 B5 B1060.2	15600
F9 B5 B1058.3	15600
F9 B5 B1051.6	15600
F9 B5 B1060.3	15600
F9 B5 B1049.7	15600

2015 Launch Records

List the records which will display the month names, failure landing_outcomes in drone ship ,booster versions, launch_site for the months in year 2015.

Note: SQLite does not support monthnames. So you need to use substr(Date, 6,2) as month to get the months and substr(Date,0,5)='2015' for year.

```
task9_query = """
SELECT
    CASE substr(Date, 6, 2)
        WHEN '01' THEN 'January'
        WHEN '02' THEN 'February'
        WHEN '03' THEN 'March'
        WHEN '04' THEN 'April'
        WHEN '05' THEN 'May'
        WHEN '06' THEN 'June'
        WHEN '07' THEN 'July'
        WHEN '08' THEN 'August'
        WHEN '09' THEN 'September'
        WHEN '10' THEN 'October'
        WHEN '11' THEN 'November'
        WHEN '12' THEN 'December'
    END AS Month,
    Landing_Outcome,
    Booster_Version,
    Launch_Site
FROM SPACEXTABLE
WHERE Landing_Outcome = 'Failure (drone ship)'
    AND substr(Date, 1, 4) = '2015'
"""

%sql $task9_query
```

```
* sqlite:///my_data1.db
Done.
```

Month	Landing_Outcome	Booster_Version	Launch_Site
January	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
April	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order.

```
task10_query = """
SELECT Landing_Outcome, COUNT(*) AS Outcome_Count
FROM SPACEXTABLE
WHERE Date BETWEEN '2010-06-04' AND '2017-03-20'
GROUP BY Landing_Outcome
ORDER BY Outcome_Count DESC
"""
```

```
%sql $task10_query
```

```
* sqlite:///my_data1.db
```

Done.

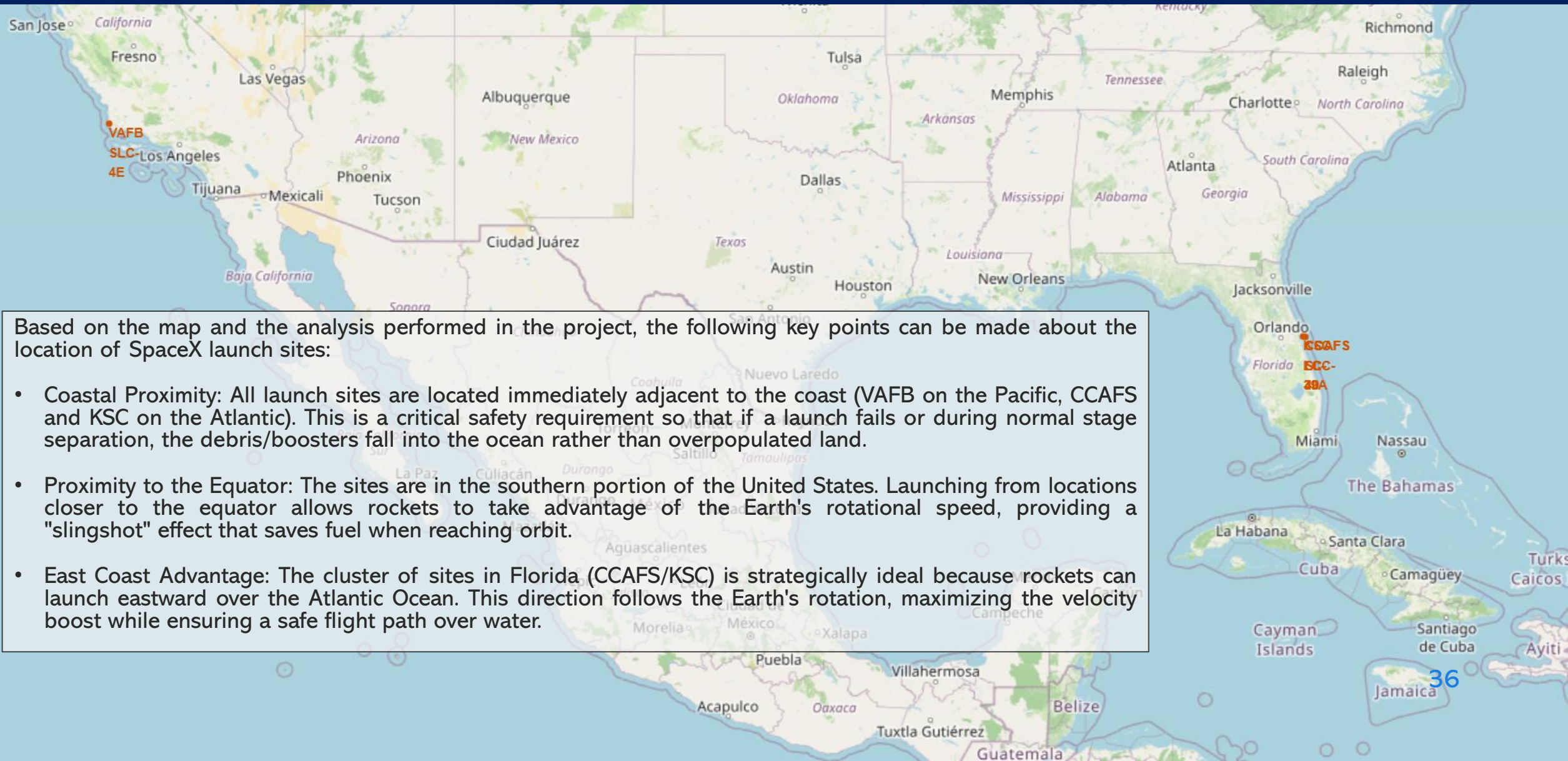
Landing_Outcome	Outcome_Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The image is a composite of a solid blue rectangle on the left and a satellite photograph of Earth on the right. The Earth shows the horizon, clouds, and glowing city lights, particularly concentrated in the lower right quadrant.

Section 3

Launch Sites Proximities Analysis

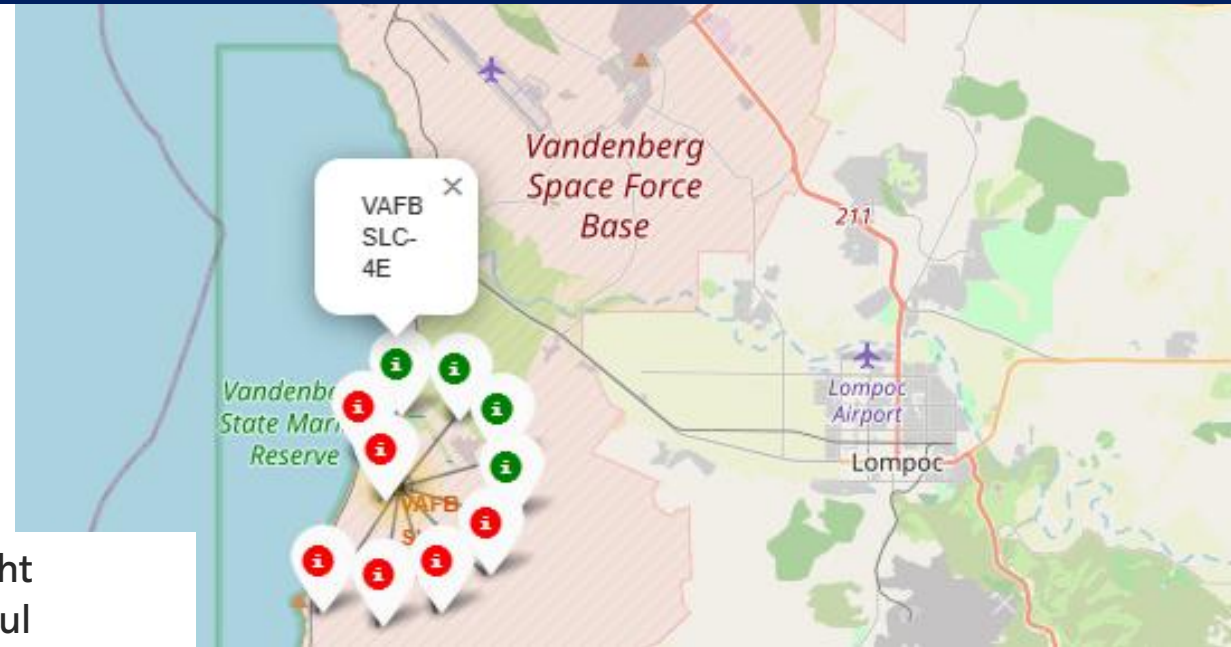
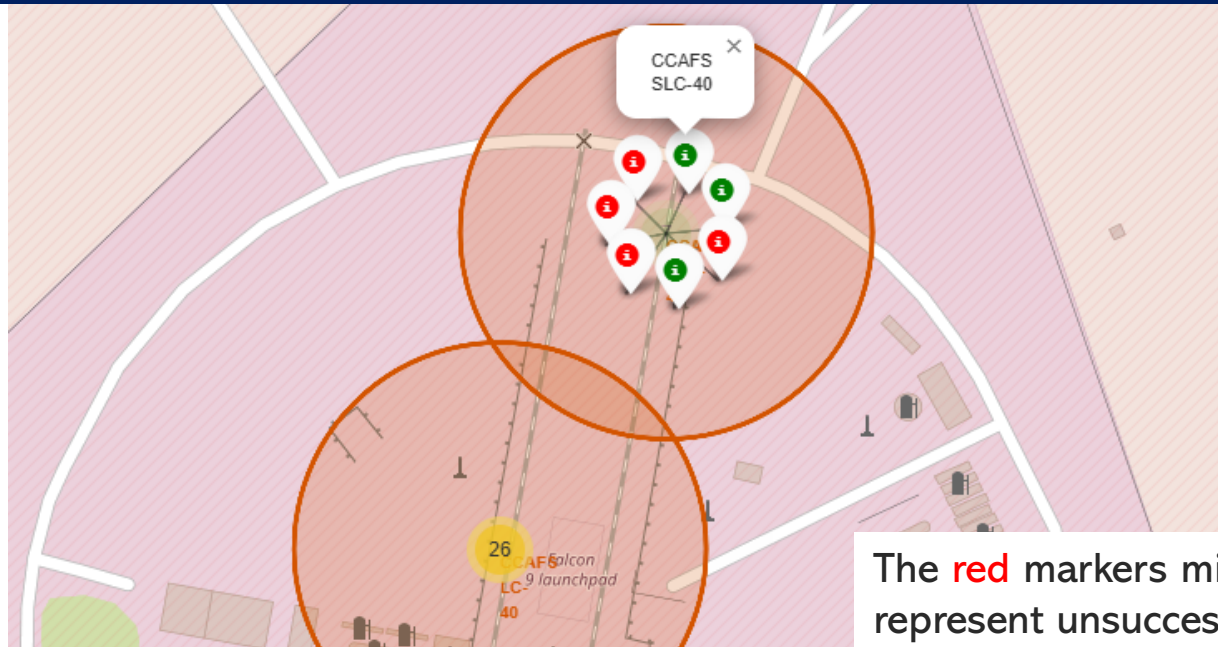
Proximity Analysis 1



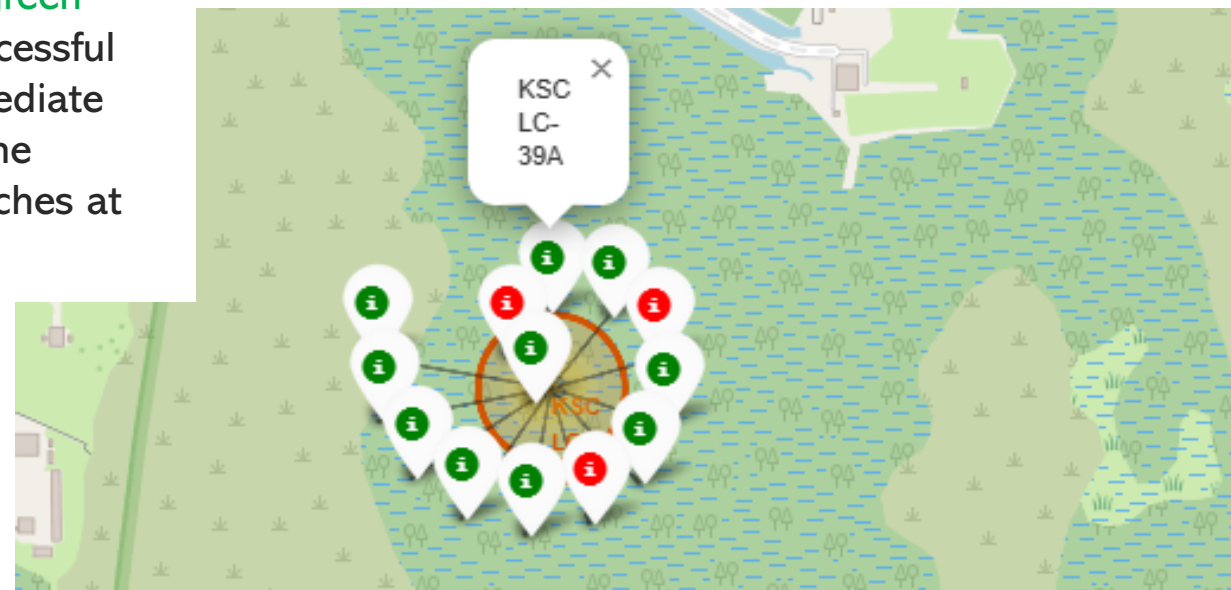
Based on the map and the analysis performed in the project, the following key points can be made about the location of SpaceX launch sites:

- **Coastal Proximity:** All launch sites are located immediately adjacent to the coast (VAFB on the Pacific, CCAFS and KSC on the Atlantic). This is a critical safety requirement so that if a launch fails or during normal stage separation, the debris/boosters fall into the ocean rather than overpopulated land.
- **Proximity to the Equator:** The sites are in the southern portion of the United States. Launching from locations closer to the equator allows rockets to take advantage of the Earth's rotational speed, providing a "slingshot" effect that saves fuel when reaching orbit.
- **East Coast Advantage:** The cluster of sites in Florida (CCAFS/KSC) is strategically ideal because rockets can launch eastward over the Atlantic Ocean. This direction follows the Earth's rotation, maximizing the velocity boost while ensuring a safe flight path over water.

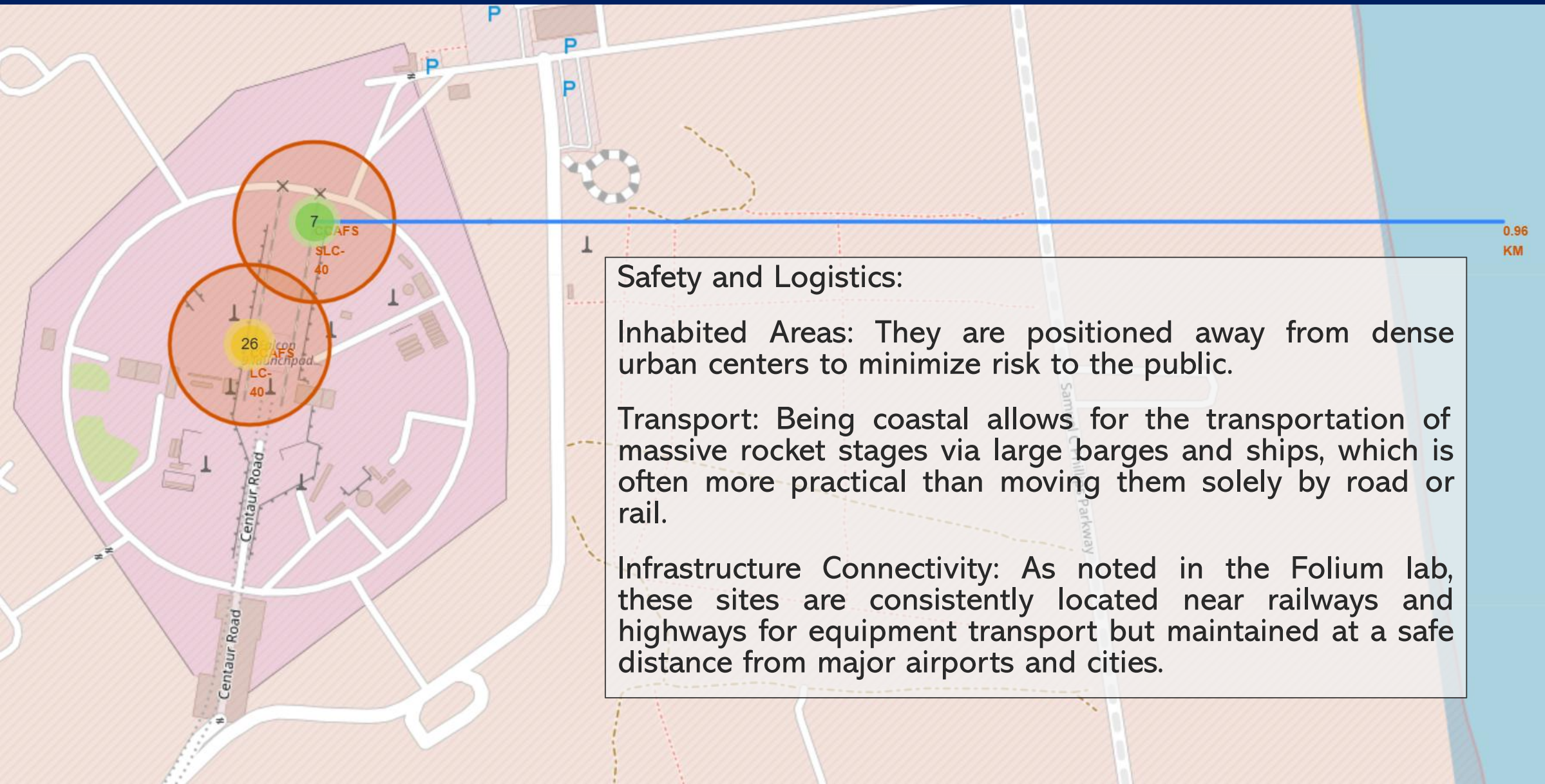
Launch outcomes for each site

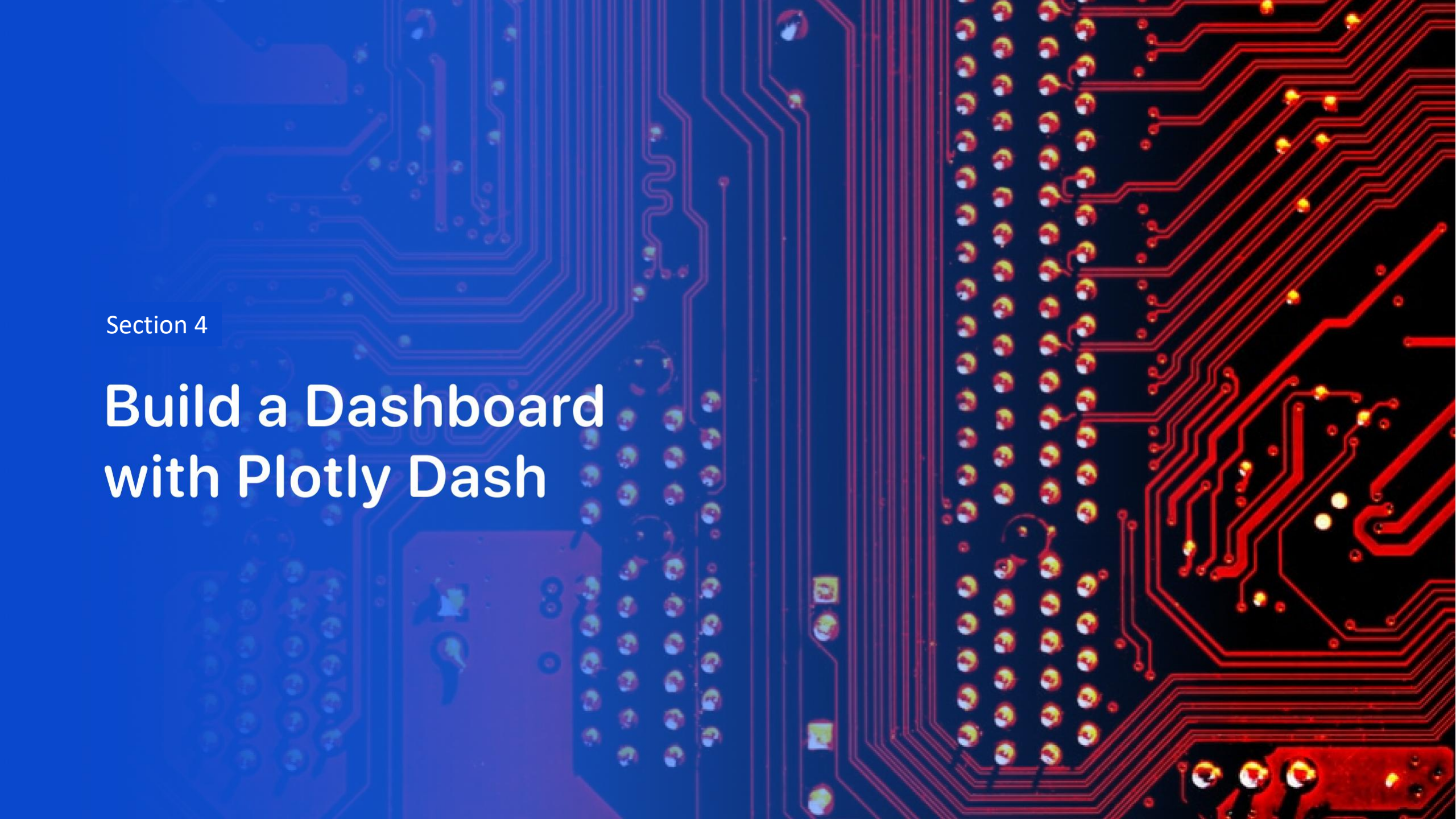


The **red** markers might represent unsuccessful launches, while the **green** markers indicate successful ones, providing immediate visual feedback on the performance of launches at each site.



Proximity Analysis 2





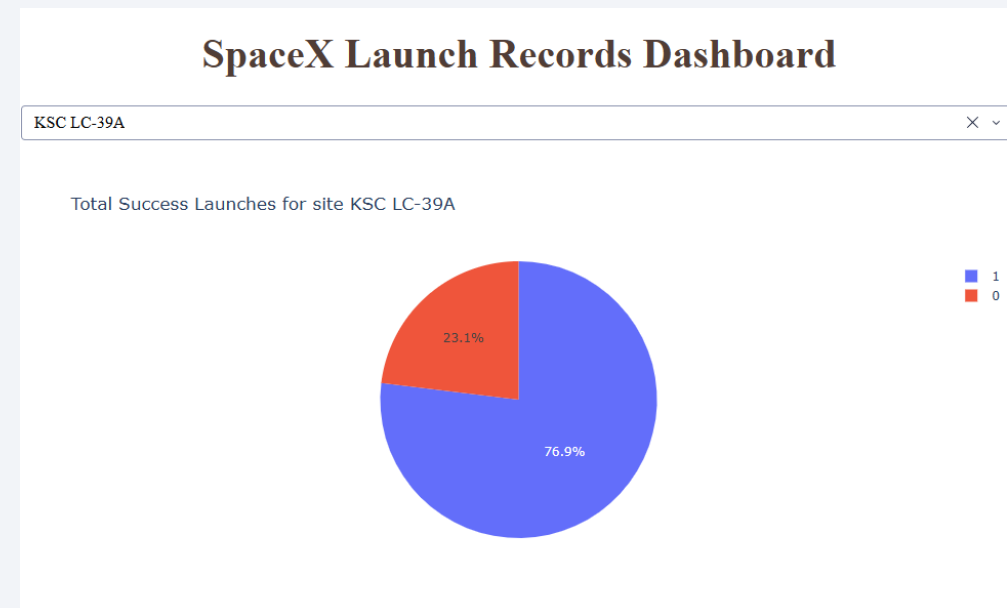
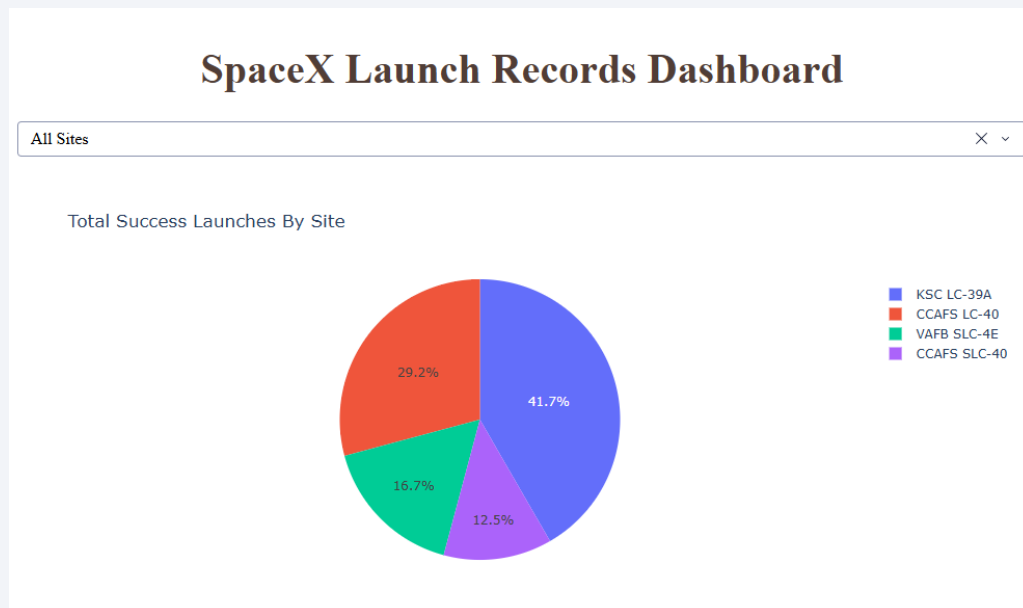
Section 4

Build a Dashboard with Plotly Dash

Key Insights from SpaceX Launch Data Dashboard

The **KSC LC-39A** launch site has the highest number of successful launches, making up **41.7%** of the total successes. This indicates that KSC LC-39A is a highly reliable site for SpaceX launches.

The significant portion of successful launches (**76.9%**) from **KSC LC-39A** highlights its reliability and effectiveness as a launch site.



Key Insights from SpaceX Launch Data Dashboard

Based on the scatter plot from the dashboard, several key insights regarding the relationship between payload mass, booster versions, and landing success can be inferred:



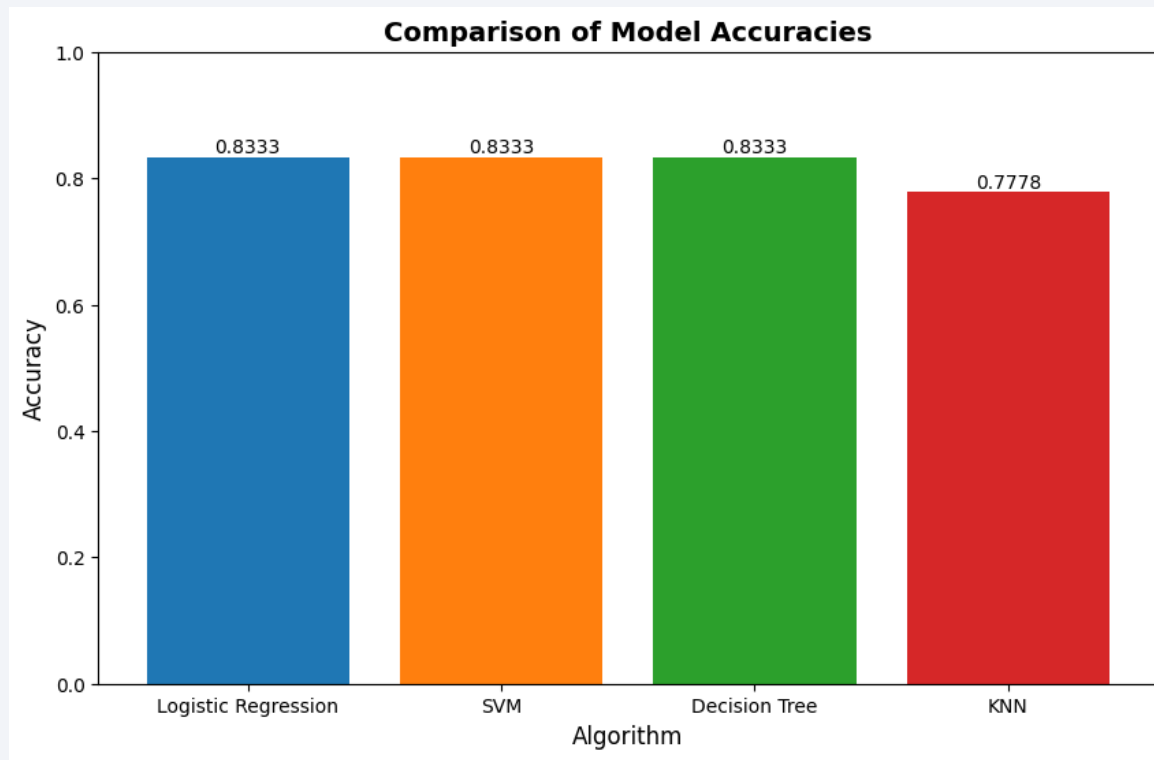
- **Booster Version Evolution & Reliability:** There is a clear correlation between the version of the booster and its success rate.
- **v1.0 (Dark Blue) and v1.1 (Red):** These older versions have a high concentration of failures (Class 0), particularly at lower payload masses.
- **FT (Full Thrust - Green):** This version shows a high density of successes (Class 1) across the 2,000kg to 6,000kg range, indicating it was a significantly more reliable platform for recovery.
- **B4 & B5 (Purple/Orange):** These newer versions show high success, including at the higher end of the payload spectrum.
- **Payload "Sweet Spot" for Success:** The highest density of successful landings (Class 1) occurs in the 2,000 kg to 4,000 kg range. This suggests that mid-range payloads provided the ideal balance of fuel and weight for a successful return.

Section 5

Predictive Analysis (Classification)

Classification Accuracy

Logistic Regression, Support Vector Machine (SVM), and Decision Tree tied for the highest performance, each achieving an accuracy score of 83.33% on the test dataset.



KNN Performance: The K-Nearest Neighbors (KNN) algorithm performed slightly worse than the others, with an accuracy of 77.78%. This indicates that a proximity-based approach is marginally less effective for this specific classification task compared to the other methods.

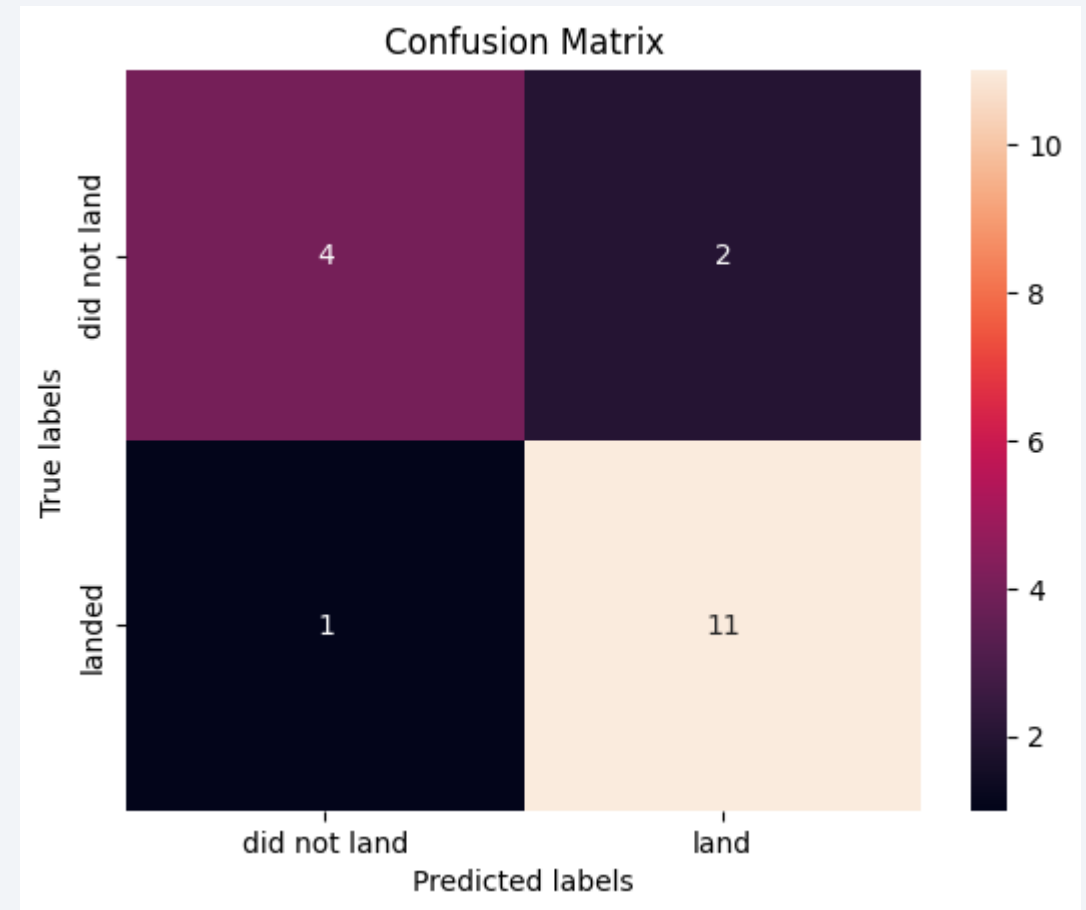
High Overall Reliability: All four models achieved scores significantly above 75%, demonstrating that the features engineered (such as Payload Mass, Orbit, and Launch Site) provide strong predictive power for determining the success of a Falcon 9 first-stage landing.

Conclusion for Deployment: Since Logistic Regression, SVM, and Decision Tree performed equally well on the test set, any of these three could be selected as the final model. However, Logistic Regression or SVM might be preferred for their relative simplicity and efficiency in binary classification tasks.

Confusion Matrix

Key Insights & Metrics (Decision Tree model)

- High Sensitivity (Recall): The model is excellent at identifying successful landings. It correctly identified 91.6% (11 out of 12) of all actual successful landings in the test set.
- Overall Accuracy: By correctly identifying 15 out of 18 cases, the model achieves the 83.33% accuracy seen in the previous bar chart.
- Minor False Alarm Rate: The main weakness of the model is its tendency to occasionally predict a "land" outcome when it actually fails (2 cases). In a real-world scenario, this "false optimism" could be risky for cost estimation.
- Reliability for "Landed" Predictions: When the model predicts a booster will land, it is correct roughly 84.6% of the time (Precision = 11 / 13).



Conclusions

Based on the comprehensive analysis across all notebooks—covering data wrangling, SQL, geographic mapping, interactive dashboards, and machine learning—here are the three most important conclusions:

1. Success is Driven by Iterative Technological Maturity

- The data reveals a direct correlation between time (Flight Number/Year) and landing success. SpaceX transitioned from a 0% success rate in the early years (2010–2013) to consistently achieving 80–90%+ success in recent years. This proves that first-stage reusability is a mature, reliable technology resulting from iterative engineering improvements rather than random chance.

2. Orbit Type and Booster Version are Critical Success Factors

- While payload mass influences the mission, the Orbit Type and Booster Version are the strongest physical predictors of a successful landing:
- Orbits: Missions to VLEO (Starlink), SSO, and ES-L1 have near-perfect success rates, whereas GTO missions (which require more fuel and higher speeds) remain more challenging.
- Boosters: The evolution from v1.0 to Block 5 shows a massive jump in reliability. Newer boosters (B4 and B5) are significantly more capable of landing successfully even when carrying maximum payload masses (~15,600 kg).

3. Landing Outcomes are Highly Predictable for Business Strategy

- The machine learning phase proved that landing outcomes are not arbitrary. With both Logistic Regression and SVM models achieving 83.3% accuracy, SpaceX can statistically predict the likelihood of recovering a booster before it even launches. This predictive capability is the foundation of their business model, allowing them to accurately price missions and significantly undercut competitors by factoring in the high probability of booster reuse.

Thank you!

